ICP and ISP Programmer Guide

2025/11/07

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Agenda

- Test Environment
- ICP Program
- Run SVF update CPLD
- ISP Program

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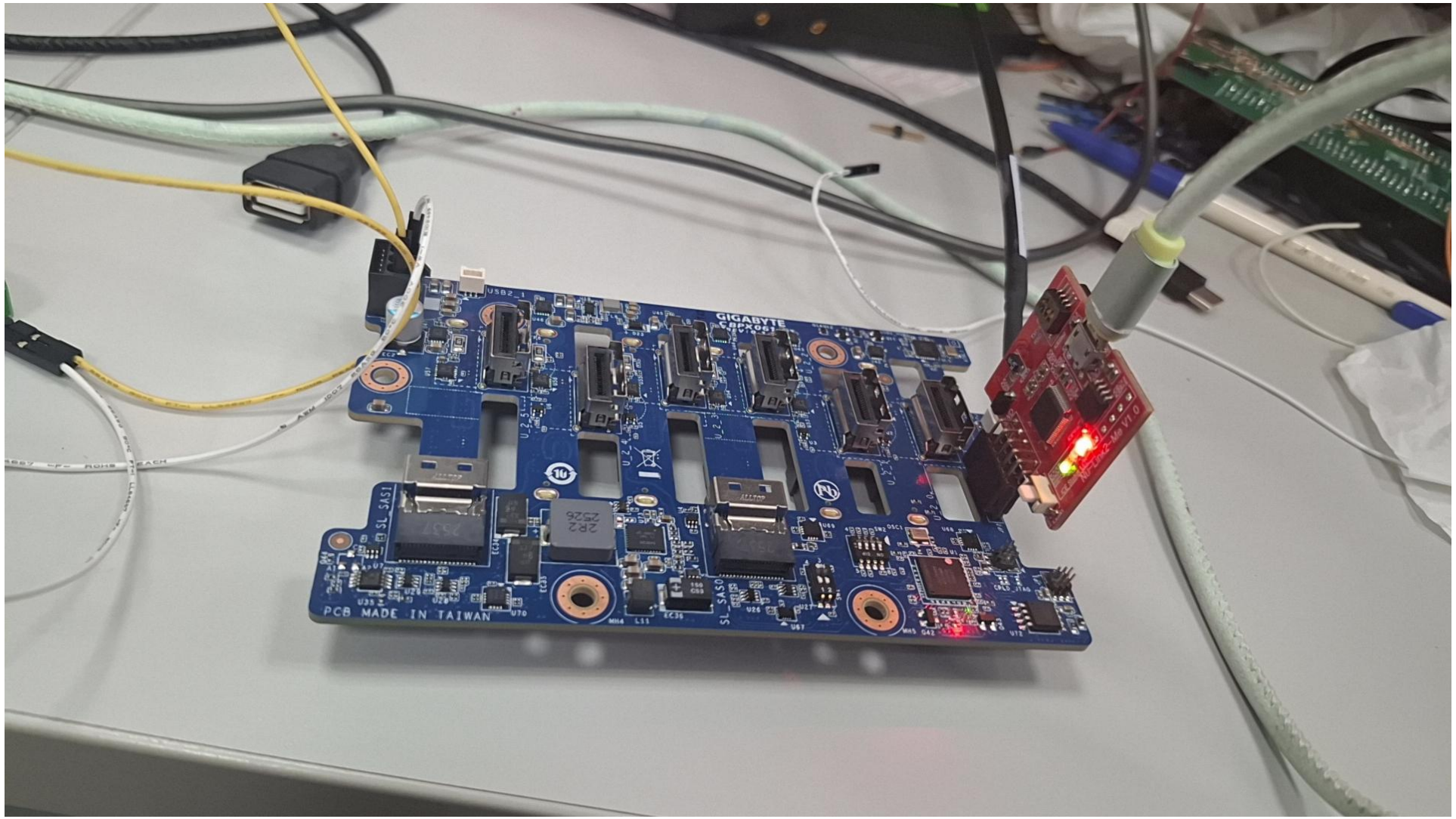
TEST Env.



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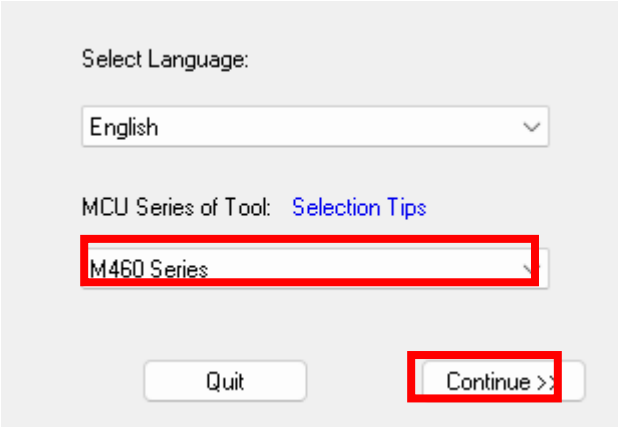
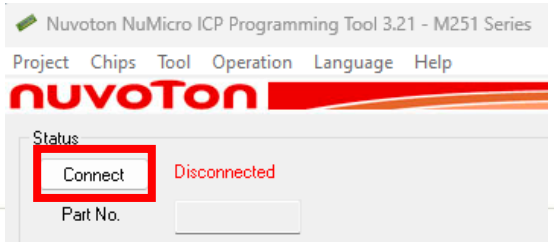
ICP Program

- Use ICP Programming Tool
  - https://www.nuvoton.com/resource-download.jsp?tp_GUID=SW1720200221181328
- Install and execute

1. Select “M460 Series” and click the “Continue” button



2. Click the “Connect” button



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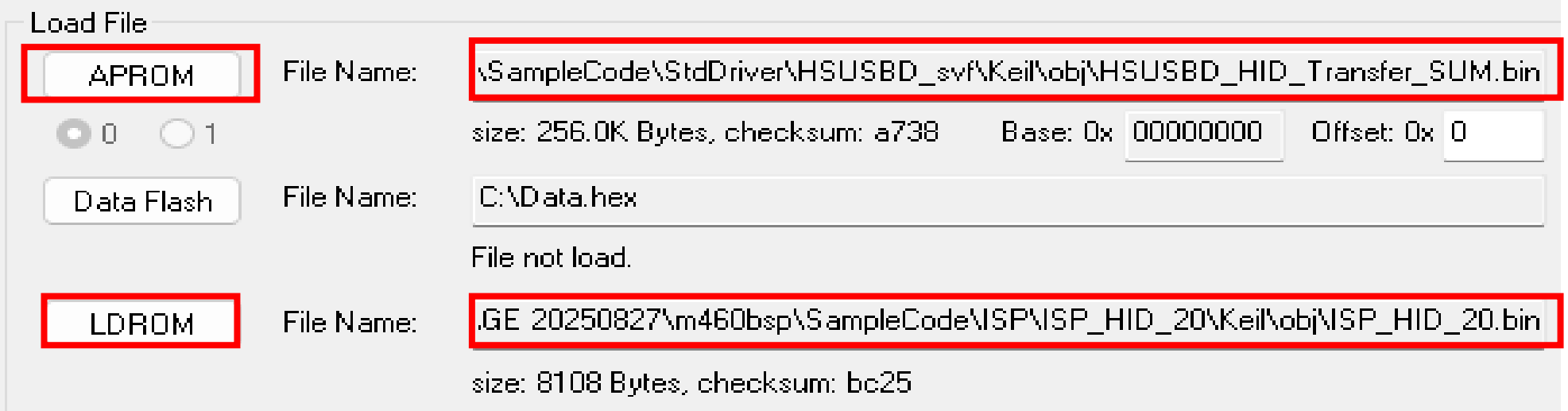
ICP Program

3. Click the “APROM” and “LDROM” button and select the target file.



APROM bin(*sum.bin)
\m460bsp\SampleCode\StdDriver\HSUSBD_svf\Keil\obj\HSUSBD_HID_Transfer_SUM.bin

LDROM bin
\m460bsp\SampleCode\ISP\ISP_HID_20\Keil\obj\ISP_HID_20.bin

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ICP Program

- Config Setting

Chip Settings

Setting | Config 0: 0xFF97FFFF | Config 1: 0xFFFFFFFF | < Update Hist

Config 2: 0xFFFF5A5A | Config 3: 0xFFFFE2FF

CONFIG 0-3 | XOM

Chip Booting Selection

- ○ APROM
- ● LDROM

I/O Initial State Selection

- ● Input Tri-state Mode
- ○ Quasi Bi-directional Mode

Brown Out Voltage

- ○ 3.0V ○ 2.8V ○ 2.6V ● 2.4V
- ○ 2.2V ○ 2.0V ○ 1.8V ○ 1.6V
- ☐ Brown Out Detector ☐ Brown Out Reset

Watchdog Timer Mode Selection

- ● WDT is inactive.
- ○ WDT is active and WDT clock is always on.
- ○ WDT is active and WDT clock is controlled by LIRCEN in power-down.

Boot Loader ISP Mode

- ☐ UART (PB.12/PB.13) ☐ USB
- ☐ CAN (PA.4/PA.5) ☐ I2C (PC.0/PC.1)
- ☐ SPI (PA.0/PA.1/PA.2/PA.3)

Data Flash Options

- ☐ Data Flash Enable
- Base Address: 0x FFFFFFFF
- Data Flash Size: 0.00K

Lock Options

- ☐ Security Lock
- Advance Security Lock: 0x 5A
- Key Store Protect Lock: 0x 5A
- ☐ Disable ICE Function

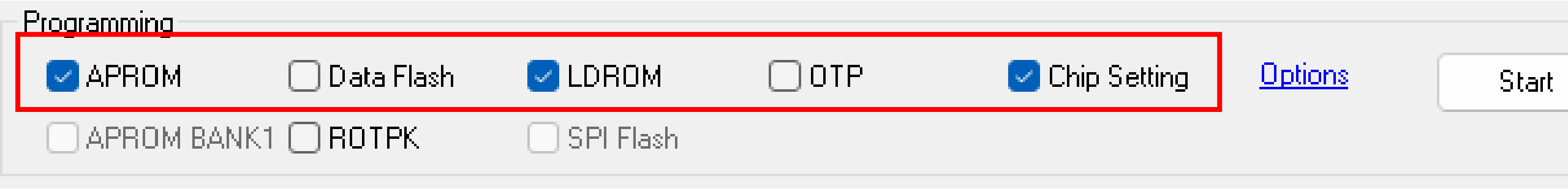
Config Value

Config 0: 0xFF9FFF7F | Config 1: 0xFFFFFFFF | Config 2: 0xFFFF5A5A | Config 3: 0xFFFFE2FF

OK | Cancel

ICP Program

Set “APROM”, “LDROM” and “Chip Setting”. And start program



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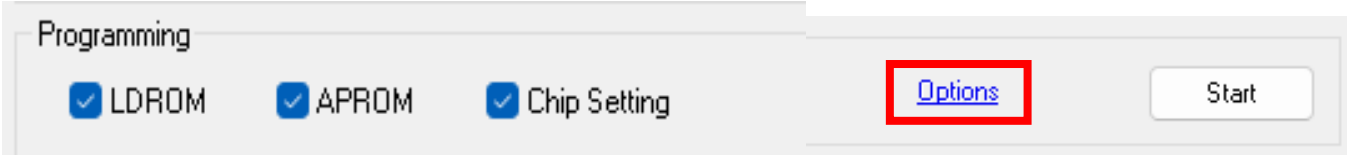
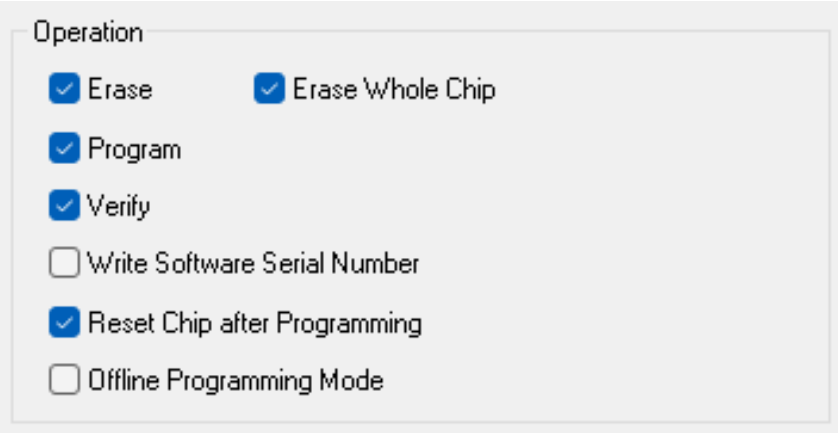
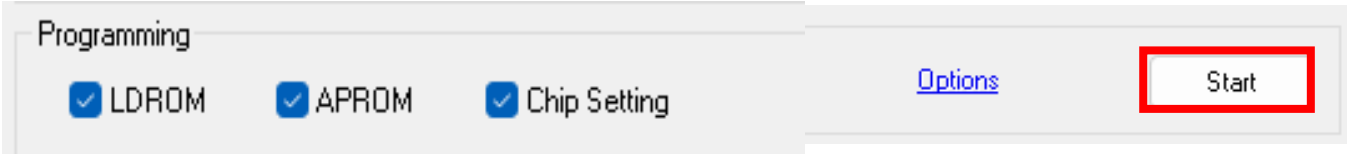
ICP Program

3. Click the “Options” and set setting shown as right.

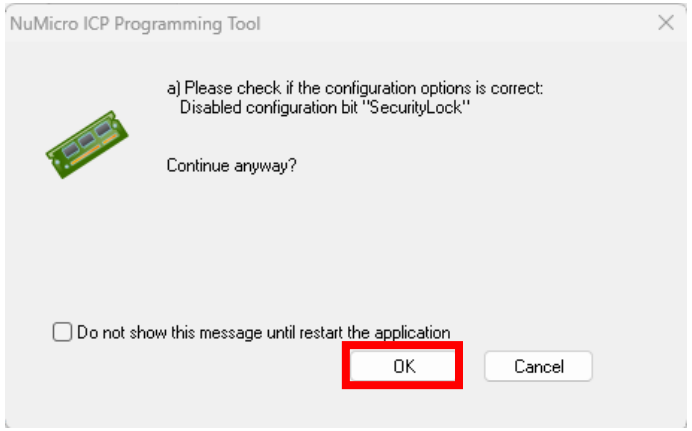


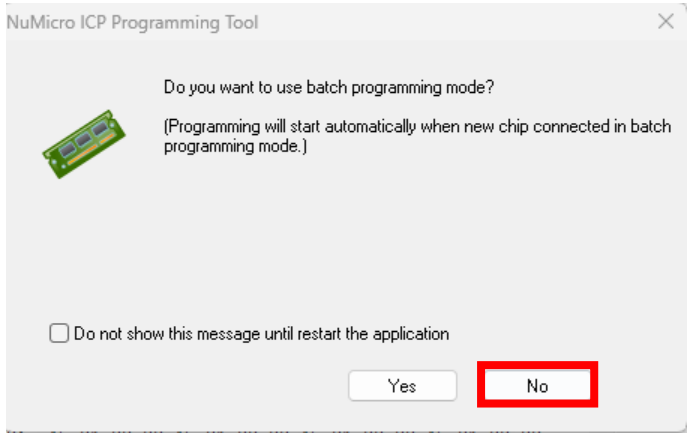


4. Click the “Start” button.



Click the “OK” and “No” button when it shows the below messages.





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Joy of innovation

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谢谢

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Děkuji

Bedankt

Thank you

Kiitos

Merci

Danke

Grazie

ありがとう

감사합니다

Dziękujemy

Obrigado

Спасибо

Gracias

Teşekkür ederim

Cảm ơn